

GHANA INSTITUTE OF MANAGEMENT AND PUBLIC ADMINISTRATION

School of Technology | Department of Computer Science

CS 305: Problem Set 4 (Undecidability & NP-Completeness)

Weeks 12–14 | Total Marks: 100 | Weight: 7.5% of Final Grade

Instructions & Guidelines

- Mapping reduction proofs ($A \leq_m B$) must clearly define the computable reduction function f and prove both directions ($w \in A \iff f(w) \in B$).
- NP-completeness proofs must prove both: (1) Membership in NP, and (2) Polynomial-time reduction from a known NP-complete problem.

Problem 1: Undecidability via Mapping Reductions (25 Marks)

Let $E_{\text{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$.

- (a) Prove that E_{TM} is ****undecidable**** by showing a mapping reduction $A_{\text{TM}} \leq_m \overline{E_{\text{TM}}}$ (or reduction from A_{TM} to E_{TM}) (Sipser Theorem 5.2). (15 Marks)
- (b) Use your result to prove that $EQ_{\text{TM}} = \{\langle M_1, M_2 \rangle \mid L(M_1) = L(M_2)\}$ is also undecidable. (10 Marks)

Problem 2: Rice's Theorem Applications (25 Marks)

Rice's Theorem states that any non-trivial semantic property of a Turing machine's language is undecidable. Using Rice's Theorem, determine (with rigorous justification) whether each of the following languages is decidable or undecidable:

- (a) $L_1 = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ contains the string } 101\}$. (8 Marks)
- (b) $L_2 = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is a context-free language}\}$. (8 Marks)
- (c) $L_3 = \{\langle M \rangle \mid M \text{ is a TM and } M \text{ has at most 5 states}\}$. (9 Marks)

Problem 3: NP-Completeness Reduction: 3SAT to CLIQUE (25 Marks)

A ****clique**** in an undirected graph $G = (V, E)$ is a subset of vertices $V' \subseteq V$ such that every pair of distinct vertices in V' is connected by an edge. Let:

$$\text{CLIQUE} = \{\langle G, k \rangle \mid G \text{ is an undirected graph containing a clique of size } k\}$$

- (a) Prove that $\text{CLIQUE} \in \mathbf{NP}$ by defining a polynomial-time verification algorithm. (8 Marks)
- (b) Given the 3CNF Boolean formula $\phi = (x_1 \vee \overline{x_2} \vee x_3) \wedge (\overline{x_1} \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee \overline{x_3})$, construct the exact graph G_ϕ and target integer k according to the Cook-Karp 3SAT-to-CLIQUE reduction (Sipser Theorem 7.32). Draw the graph G_ϕ . (10 Marks)
- (c) Prove that ϕ is satisfiable $\iff G_\phi$ contains a clique of size $k = 3$. (7 Marks)

Problem 4: Modern Applied Case Study — Ethereum Smart Contract Gas (25 Marks)

The Ethereum Virtual Machine (EVM) executes Turing-complete bytecode submitted by untrusted network users.

- (a) Prove why it is mathematically impossible for an Ethereum miner to statically pre-screen smart contracts to reject all transactions containing infinite loops before execution begins. (12 Marks)
- (b) Formulate how Ethereum's deterministic Gas metering mechanism solves the practical consequences of Turing's Halting Problem, guaranteeing bounded execution time for network consensus. (13 Marks)